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Prediction of the Groundwater Levels Based on Random Forest Regression Algorithm

Pengbo Wang[1*]
1. Shanghai Zhongqiao Vocational and Technical University, Shanghai, 201514, China
Corresponding Author: Pengbo Wang Email: wangpb@shzq.edu.cn

***Abstract*—Random forest regression is a widely used machine learning algorithm. In this study, random forest regression is employed to predict groundwater levels. Five influencing factors are considered: river flow, temperature, saturation deficit, precipitation, and evaporation capacity. Each factor is regarded as a feature. Twenty-four sets of data were collected: eighteen sets for training and six for testing. The groundwater levels predicted using the proposed approach agree well with the corresponding true values. The relative errors are less than 5%, which satisfies the requirements of engineering precision. A feature importance analysis is also conducted. The precipitation has the most significant influence on the groundwater levels, with a degree of more than 50%, and the saturation deficit has the least influence, less than 10%. The results of this study not only provide an effective approach for groundwater level prediction but also demonstrate the encouraging prospects of machine learning algorithms in hydrological engineering applications.**

Keywords*—machine learning; groundwater level; random forest; regression; precipitation

I. INTRODUCTION

To utilize and manage the groundwater resources properly, it is necessary to give an accurate prediction of the groundwater levels. However, groundwater systems are highly nonlinear and complex. The groundwater level is influenced by many factors, such as temperature, precipitation, and evaporation [1]. In many cases, accurately forecasting groundwater levels is challenging. Many researchers have used differential equations to address this issue [2~4]. Nevertheless, the variation of the groundwater level is a complicated nonlinear process affected by many factors, and the differential equation is an idealized model that cannot appropriately consider each factor. The establishment of an accurate differential equation requires a large number of hydrological data, which is not always feasible in practice [5].

Machine learning algorithms are mathematical models that enable computers to make judgements or predictions based on data sets without explicit programming [6]. Among these algorithms, an effective one is the random forest, which can be used for both classification and regression tasks with a high degree of accuracy [7]. Random forest regression possesses a remarkable nonlinear functional mapping capability and can determine the feature importance of each factor directly [8]. The random forest algorithm is widely used in various industries, such as fraud detection in banking, disease diagnosis in healthcare, price prediction in stocks, and customer segmentation in e-commerce. In this study, the random forest regression algorithm is employed to predict the groundwater levels.

The remainder of this article is organized as follows. In the second section, the mechanism of the random forest regression algorithm is introduced. In the third section, the data sets, including influencing factors and groundwater levels, are described. In the fourth section, the prediction accuracy of the proposed method is verified. In the fifth section, the feature importance of each influencing factor is discussed. Conclusions are presented in the final section.

II. RANDOM FOREST REGRESSION

Regression is a mathematical approach used to predict numerical values over a certain range. The random forest regression algorithm is an ensemble learning method that comprises multiple decision trees. The steps of the algorithm are as follows.

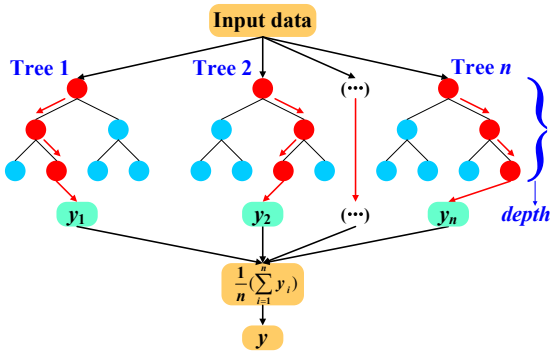


Fig. 1. Topology structure of random forest regression algorithm.

(1) Input data split: The input data are divided into subsets and transferred to each tree randomly. This step can reduce the similarity among the trees and enhance the robustness of the forest.

(2) Decision tree training: Each decision tree is trained using its corresponding data subset. In many cases, the number of trees is large, and this step can be performed in parallel.

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(3) Aggregating for final output: The predictions of all the trees are aggregated, and their average value is considered to be the final output prediction. This step can improve the predicting stability and accuracy.

The final prediction of the random forest regression algorithm is expressed as

$$y = \frac{1}{n}(\sum_{i=1}^{n} y_i) \quad (1)$$

where n is the number of decision trees, y_i is the prediction corresponding to the ith decision tree, and y is the final prediction of the forest.

The mechanism of the random forest regression algorithm is illustrated in Fig. 1. The algorithm can be implemented efficiently and effectively using the scikit-learn framework in the Python programming environment.

III. Data Sets

TABLE I. FACTORS INFLUENCING THE GROUNDWATER LEVELS IN THE TRAINING SET

No.	x_{rf}	x_t	x_{sd}	x_p	x_{ec}	y_{gl}
1	10.0	23.0	8.0	29.0	5.6	6.63
2	9.0	21.0	6.0	74.0	4.6	5.62
3	7.0	15.0	5.0	21.0	2.3	5.96
4	9.5	8.5	5.0	15.0	3.5	6.30
5	5.5	0	6.2	14.0	2.4	6.80
6	12.0	-8.5	4.5	11.0	0.8	6.90
7	1.5	-11.0	2.0	1.0	1.3	6.70
8	3.0	-7.0	2.5	2.0	1.3	6.77
9	7.0	0	3.0	4.0	4.1	6.67
10	19.0	10.0	7.0	0	3.2	6.33
11	4.5	18.0	10.0	19.0	6.5	5.82
12	8.0	21.5	11.0	81.0	7.7	5.58
13	57.0	22.0	5.5	186.0	5.5	5.48
14	35.0	19.0	5.0	114.0	4.6	5.38
15	39.0	13.0	5.0	60.0	3.6	5.51
16	23.0	6.0	3.0	35.0	2.6	5.84
17	11.0	1.0	2.0	4.0	1.7	6.32
18	4.5	-7.0	1.0	6.0	1.0	6.56

The collected data sets are divided into a training set and a testing set. The former is used by the model to learn from data samples to yield an outcome, and the latter is used to verify the accuracy of the model. Generally, the number of samples in the testing set accounts for 20%~25% of the total data samples available.

A. Training Set

In this study, the training set comprises 18 sets of data, as listed in Table I. Five influencing factors are considered for the effect of the groundwater level. Each factor is regarded as a feature in the algorithm.

The meanings of the symbols are as follows:

x_{rf}: river flow; x_t: temperature;

x_{sd}: saturation deficit; x_p: precipitation;

x_{ec}: evaporation capacity; y_{gl}: groundwater level.

B. Testing Set

The testing set comprises 6 sets of data, as listed in Table II. The meanings of the symbols are the same as in Table I.

TABLE II. FACTORS INFLUENCING THE GROUNDWATER LEVELS IN THE TESTING SET

No.	x_{rf}	x_t	x_{sd}	x_p	x_{ec}	y_{gl}
1	1.5	-10.0	1.2	1.0	1.2	6.92
2	1.8	-10.0	2.0	1.0	0.8	6.97
3	4.0	-2.0	2.5	6.0	2.4	6.84
4	13.0	10.0	5.0	30.0	4.4	6.50
5	5.0	17.0	9.0	18.0	6.3	5.75
6	9.0	22.0	10.0	113.0	6.6	5.54

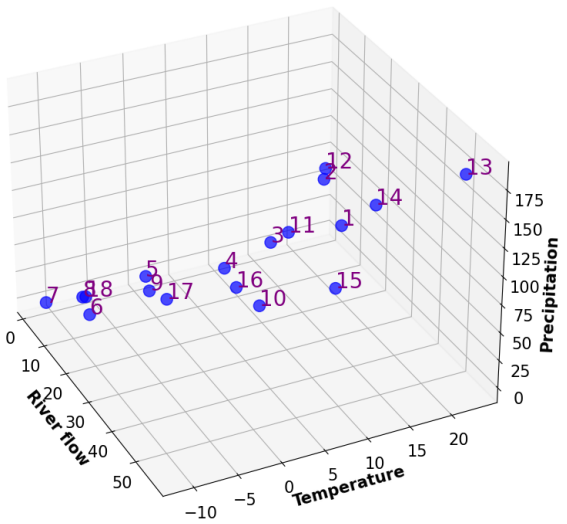


Fig. 2. Distribution of training set data with respect to different values of river flow, temperature, and precipitation.

The spatial distributions of the training set data and testing set data with respect to different values of river flow, temperature, and precipitation are shown in Figs. 2 and 3, respectively. A three-dimensional plot can display only three components; therefore, the distributions of the saturation deficit and evaporation capacity are not shown.

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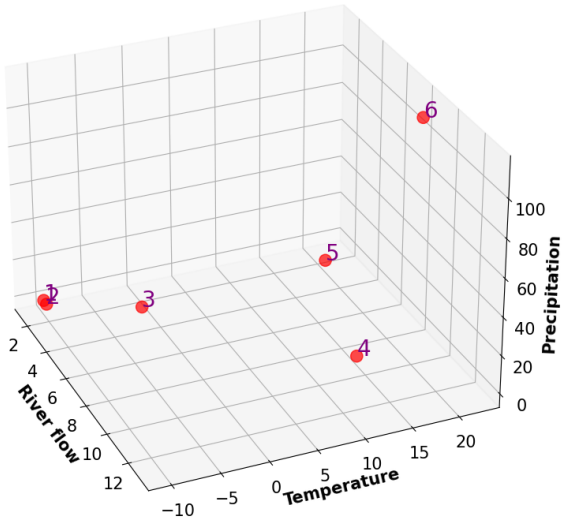


Fig. 3. Distribution of testing set data with respect to different values of river flow, temperature, and precipitation.

IV. Prediction Accuracy

Hyperparameters have immense importance for the performance of machine learning algorithms. They are external to the model and cannot be evaluated using the given data [9]. No general rule exists for determining the optimal value for each specific problem. In practice, the hyperparameters can be estimated using rules of thumb, values copied from other issues, or parameter tuning.

In the random forest regression algorithm, two hyperparameters have significant effects on the final output: the number of trees in the forest and the depth of the tree [10]. To derive the appropriate hyperparameter values, the mean squared error (MSE), which is widely used in scientific computing, is adopted. The MSE is used to assess the average squared difference between the predicted and true values, expressed as:

$$MSE = \frac{1}{K}\sum_{j=1}^{K}(y_{j,pred} - y_{j,true})^2 \qquad (2)$$

where K represents the number of samples in the data set, $y_{j,pred}$ represents the predicted value of the proposed algorithm corresponding to the jth sample, and $y_{j,true}$ represents the true value.

Next, Equation (2) is applied to the training set. The MSE values for the training set with respect to different values of n and d are listed in Table III, where n is the number of trees in the forest, and d is the depth of the tree. The MSE values are also plotted in a heatmap, which is a graphical representation of the data where the matrix values are depicted by color for quick intuitive comprehension, as shown in Fig. 4. For n = 7 and d = 6, the MSE reaches its minimum. Consequently, in this study, n = 7 and d = 6 are treated as the best values for the forest layout.

TABLE III. MSE VALUES FOR THE TRAINING SET (10^{-3})

MSE	$d=3$	$d=4$	$d=5$	$d=6$	$d=7$	$d=8$
$n=5$	14.72	10.92	11.10	10.99	10.99	10.99
$n=6$	12.27	8.53	8.57	8.46	8.46	8.46
$n=7$	12.11	7.86	7.74	7.66	7.66	7.66
$n=8$	12.77	7.97	7.89	7.84	7.84	7.84
$n=9$	14.99	9.69	9.63	9.58	9.58	9.58
n=10	16.69	10.79	11.24	11.21	11.21	11.21
n=11	16.14	10.71	11.03	11.00	11.00	11.00
n=12	19.08	13.70	14.03	14.00	14.00	14.00

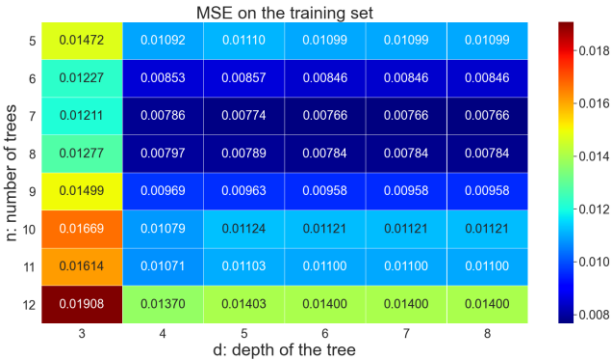


Fig. 4. MSE values of the training set with respect to different values of n (number of trees) and d (depth of the tree).

With the random forest configuration determined, the groundwater levels can be predicted using the proposed algorithm. To verify its performance, the index of relative error is introduced, which is defined as:

$$relative\ error = \frac{|y_{gl,pred} - y_{gl,true}|}{y_{gl,true}} \times 100\% \qquad (3)$$

where $y_{gl,pred}$ is the groundwater level predicted using the proposed algorithm with the factor values in the testing set, and $y_{gl,true}$ is the corresponding true level in the testing set.

TABLE IV. PREDICTION ACCURACY OF THE TESTING SET

No.	$y_{gl,true}$	$y_{gl,pred}$	Relative Error
1	6.920	6.719	2.91%
2	6.970	6.853	1.68%
3	6.840	6.497	5.01%
4	6.500	6.201	4.59%
5	5.750	5.929	3.11%
6	5.540	5.534	0.10%

The relative errors are calculated and are listed in Table IV. The groundwater levels predicted using the random forest regression algorithm agree well with the true values, and the relative errors are less than 5%, which

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is sufficiently accurate for most hydrological engineering applications.

V. Feature Importance

To study the extent of the impact of each influencing factor on groundwater levels, the concept of feature importance is introduced, which reveals the contribution of each feature to the performance of the model. In this study, influencing factors are perceived as features. A forest is composed of multiple decision trees, each of which comprises a group of internal nodes and leaves. Each time a feature is selected for splitting, the value of how it decreases the MSE on average is collected. The average of all trees in the forest is regarded as the criterion for feature importance. This approach can be implemented during the training process using the scikit-learn framework in the Python programming environment.

TABLE V. Feature Importance Values

No.	Influencing factor	Feature importance
1	River flow	13.78%
2	Temperature	20.73%
3	Saturation deficit	6.44%
4	Precipitation	50.94%
5	Evaporation capacity	8.11%

The feature importance values with respect to the different influencing factors are listed in Table V and illustrated using the bar chart in Fig. 5. The precipitation has the most significant influence on the groundwater levels, with a degree greater than 50%, and the saturation deficit has the least influence, less than 10%.

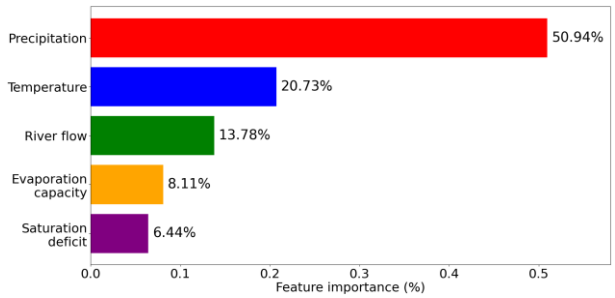


Fig. 5. Bar chart of feature importance values.

VI. Conclusion

In this study, the random forest regression algorithm is employed to predict groundwater levels. Five influencing factors are considered in this study. Eighteen sets of data in the training set and six sets in the testing set are used. The predicted groundwater levels agree well with the true values, and the relative errors are less than 5%. Feature importance is also analyzed. The results indicate that the most influential factor is precipitation, with a degree greater than 50%, and the least influential factor is the saturation deficit. This work not only provides a new method of groundwater level prediction but also demonstrates the promising prospects of machine learning algorithms for hydrological engineering applications.

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